CHAPTER 15

Synthesis Research

Reviews and meta-analyses gather all prior publications on a specific topic and provide an integrated summary of them.

15.1 Overview

Most scientific research projects seek to identify new findings derived from a single study population, but the goal of tertiary analyses is to engage in the scholarship of integration. **Synthesis research** integrates existing knowledge from previous research projects. The common types of synthesis research in the health sciences include narrative reviews, systematic reviews, and meta-analyses (**Figure 15-1**). These types of research summarize what is already known about a topic, show the connections among previous studies, and offer new interpretations of previous studies' contributions to scientific knowledge. A review article in the health sciences requires:

- An extensive search of the literature
- The extraction of key information from relevant articles
- The clear and concise presentation of this information

Conducting synthesis research is one way to become an expert in the literature on a well-defined topic. This knowledge is a good outcome in and of itself, and a tertiary analysis can also be a helpful step in preparing for future primary or secondary analyses. Well-written and comprehensive review articles often become foundational for new research in the field because they summarize what is known about an area of inquiry. Because reviews provide a concise summary of the literature, published review articles may be cited more frequently than the typical article reporting the results of a primary or secondary analysis.

There also some limitations associated with synthesis research. Not all journals publish review articles, especially reviews that the editors do not solicit, so the likelihood of publication might be lower for tertiary studies than for other study approaches. Also, reviews are sometimes regarded as exhibiting less originality than other types of scholarship. A good review requires meticulous library work followed by the careful compilation and interpretation of scientific information, yet reviews are sometimes perceived to be a less rigorous form of research than projects collecting new data or involving statistical analysis.

FIGURE 15-1 Key Characteristics of Reviews and Meta-analyses

Approach	Narrative Review	Systematic Review	Meta-analysis
Objective	Synthesize existing knowledge	Synthesize existing knowledge	Synthesize existing knowledge
Primary study question	What conclusions about this topic are supported by previous studies?	When all previously published studies on this topic are examined, what conclusions can be drawn?	When the results of all previously published studies on this topic are merged, what is the summary statistic?
Population	Published literature	Published literature	Published literature
When to use the approach	The goal is to describe a new perspective on a topic that can be supported by the existing literature.	The goal is to compare the findings of previous studies on a well-defined topic.	The goal is to summarize previous findings using pooled statistics.
Requirements	The researcher has excellent library access.	The researcher has excellent library access.	The researcher has excellent library access.
	The researcher has a unique perspective on the topic.	The researcher can obtain every relevant article.	The researcher has strong quantitative skills.
First steps	Decide what key message the article will convey.	1. Decide on the specific objectives of the review. 2. Select the search methods that will be used to find potentially relevant articles. 3. Select inclusion and exclusion criteria for articles.	1. Decide on the specific objectives of the review. 2. Select the search methods that will be used to find potentially relevant articles. 3. Select the inclusion and exclusion criteria for the articles. 4. Decide how to assess the quality of the studies. 5. Decide how the results of the studies will be combined into one summary statistic.
What to watch out for	Limited publication venues	Publication bias	Studies that cannot be fairly compared
Key statistical measure	No statistics are required.	No statistics are required, but reporting some results from included studies may be helpful.	Summary measures for included studies must be reported.

15.2 Selecting a Topic

When starting a tertiary analysis, the most important decision is the selection of a topic that is narrow enough that all the relevant publications can be acquired. The topic may need to be modified if a preliminary literature search does not yield an appropriate number of articles. If an initial search of an abstract database yields only 8 possibly relevant articles, the topic probably needs to be expanded. If a search produces 352 articles, the topic needs to be narrowed to a more specific disease condition, a smaller geographic area, or a reduced scope. For example, a review of risk factors for cardiovascular disease would be cumbersome. A very long book would be required to cover all the identified risk factors, and an article-length summary would provide such a superficial level of information that it would not be useful. There is a greater likelihood of success for a review article with a narrower scope—one that limits the types of risk factors, the particular cardiovascular diseases, and the population groups included in the analysis.

15.3 Library Access

No review article can be written without excellent library access because *every* relevant article must be identified and obtained during a systematic review. The article acquisition process usually requires access to a university library that allows affiliates to make numerous interlibrary loan requests. Before starting a review project, a researcher should check with a university librarian about the library's journal access policies and the fees that users may have to pay to access articles that are not part of the library's collections or subscription services. The researcher must also prepare to maintain a meticulous system for tracking articles that have already been acquired, those that have been requested but not yet received, and those that need to be requested.

15.4 Narrative Reviews

A **narrative review** provides a unique perspective about a topic by using evidence from the literature to support the author's commentary. A narrative review might summarize important clinical aspects of a disease or summarize the epidemiological profile of a well-defined population. Because they are intended to convey a perspective and not merely compile facts, narrative reviews must be carefully organized by theme, methodology, chronology, or some other guiding principle. A narrative may also be appropriate when the researcher has developed a unique conceptual framework or theory that can be illustrated with examples from the literature. However, narrative reviews are becoming less common as editors and reviewers push for the use of systematic methods. Researchers must be prepared to justify their selection of a narrative approach. A narrative review works best when the researcher has a unique perspective on a topic and a particular expertise in the field that can be drawn on without using a systematic search strategy.

15.5 Systematic Reviews

A **systematic review** uses a predetermined and comprehensive searching and screening method to identify relevant articles. This process is designed to minimize the bias that might occur when researchers handpick the articles they want to highlight. After the identification of a focused study question, the most important decisions in a systematic review are the selection of keywords and inclusion criteria. The goal is to craft a search strategy that identifies all the articles ever published on the narrow, well-defined area covered by the review. Once potentially relevant articles have been identified from a search of one or several abstract databases, each candidate article is screened to see whether it meets all of the inclusion criteria. Relevant information is extracted from all eligible articles and presented in a summary table, then the trends and key observations are summarized. In sum, the systematic review process requires:

- Identification of an appropriately narrow study question
- Selection of a well-defined and valid search strategy
- Screening of all potentially relevant articles to determine whether they meet the predefined eligibility criteria
- Extraction of relevant information from all eligible articles
- Summarization of the findings of these articles

15.6 Meta-analysis

Meta-analysis is the calculation of a pooled statistic that combines the results of similar studies identified during a systematic review. The values included in the calculation of a summary statistic should come from high-quality quantitative studies that used similar methods to collect and analyze their data. The typical steps of a meta-analysis are:

- Define the study question and develop a study protocol
- Use a comprehensive systematic search strategy to identify every possibly eligible article
- Use predefined inclusion and exclusion criteria to identify candidates for inclusion in the meta-analysis
- Extract statistical results from each of the candidate studies
- Assess the quality and comparability of each candidate study, eliminating studies that do not meet predefined inclusion requirements
- Combine comparable statistical results into one summary statistic that adjusts for the sample sizes or confidence intervals of the contributing statistical measures

The inclusion criteria for meta-analyses are usually more restrictive than they are for general systematic reviews, because a summary statistic is meaningful only when every study included in the meta-analysis has very similar definitions for exposures and outcomes as well as similar populations, study designs, and research methods. Trying to combine dissimilar studies could hide real and meaningful differences among the study populations.

15.7 Meta-synthesis

A **meta-synthesis** is a tertiary analysis that integrates the results from several different qualitative studies. A meta-synthesis might take the form of a meta-ethnography, a meta-narrative, a thematic synthesis, or another type of process. The goal of a meta-synthesis is not to create some sort of aggregate measurement, but to enhance understanding of a particular phenomenon. A meta-synthesis might look for ways that the concepts in one study relate to the concepts developed from other studies, identify the ways that the concepts in different studies are different, or propose a new interpretation of a phenomenon that draws on several previous analyses. The methods for finding relevant studies and coding them, the key results of the analysis, and the interpretation of the meta-synthesis are included in the research report.